

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	PEP8-style 4-space indentation
2	Proper File Structure	Files and folders are logically organized	Yes	Separate modules/, data/, reports/ folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. semantic_score, pause_ratio
4	Function / Method Names	Functions are clearly named	Yes	e.g. evaluate_explanation(), compute_similarity()
5	Code Comments	Inline/docstring comments used appropriately	Yes	e.g. evaluate_explanation(), compute_similarity()
6	Modular Design	Code split into reusable function/class units	Yes	One module per responsibility
7	No Redundant Code	Duplicated/unused code removed	Yes	Shared logic centralized in pipeline.py
8	Error Handling	Exceptions are handled gracefully	Yes	Validates missing API key, unknown concept names

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	transcription.py	Python (Whisper)	Called by pipeline.py and app.py	High
2	semantic_analysis.py	Python (Sentence-BERT)	Python (Sentence-BERT)	High
3	audio_features.py	Python (Librosa)	Called by pipeline.py	High
4	scoring.py	Python	Called by pipeline.py	High
5	feedback_generator.py	Python (Gemini API)	Called by app.py and report_generator.py	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean modular folder structure
Readability	5	Clear naming, docstrings throughout
Reusability	5	Each AI module independently callable
Documentation / Comments	4	Core functions documented; minor inline comments could be expanded
Overall Score	4	Key failure points handled; could add more granular exception messages